

Appendix: Extended Theoretical Analysis and Empirical Validation

In this appendix, we provide a deeper investigation into the theoretical motivations behind SPECTRA. We empirically validate the **Periodic Manifold Hypothesis** and demonstrate why our **Spectral Extraction** mechanism is distinct from and superior to naive sparsity methods used in prior works like SparseTSF - or standard RAFT.

A. Periodicity Detection: Frequency vs. Time Domain

A critical component of SPECTRA is selecting the optimal stride S and kernel size T . While prior efficient methods like SparseTSF rely on time-domain **Autocorrelation (ACF)**, we leverage **Frequency-Domain Analysis** to ground our sparsity in signal processing theory.

Power Spectral Density (PSD) Analysis

Time-domain correlation often suffers from decay over long horizons and harmonic ambiguity (the “echo” effect). In contrast, Fast Fourier Transform (FFT) reveals the global, deterministic energy distribution of the signal.

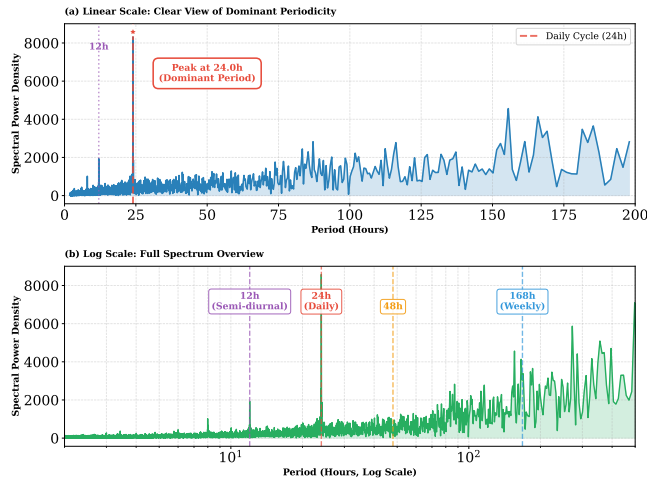


Figure 1: Power Spectral Density (PSD) Analysis. (a) **Linear Scale:** A dominant energy spike is observed strictly at $T = 24h$, validating our choice of stride $S = 24$. (b) **Log Scale:** The spectrum reveals a clear harmonic structure (12h, 24h, 168h). Unlike autocorrelation which decays due to non-stationarity, spectral density provides a robust definition of the attractor’s period.

Analysis: As illustrated in Figure 1, the dataset exhibits a sharp spectral peak at the fundamental frequency f_{dom} . We derive the motif length $T = 1/f_{dom}$. By setting our stride $S \approx T$, we align our retrieval mechanism with the dataset’s maximum information density. This ensures that the sparse dictionary covers the phase space of the attractor without aliasing, a guarantee that random subsampling cannot provide.

B. The “Spectral Extractor” as a Learned Matched Filter

We define our encoder as a “Spectral Extractor” because it functions as a learnable band-pass filter. To validate this claim, we analyzed the weights of the Conv1D layer after training.

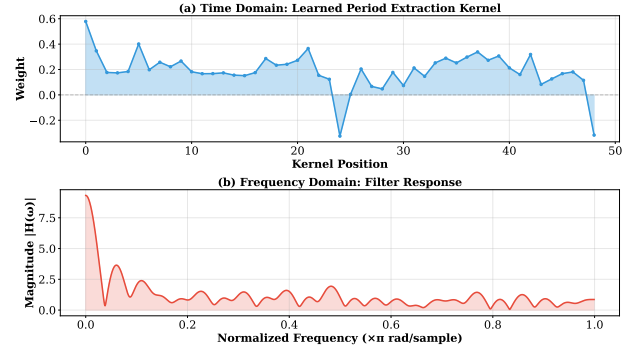


Figure 2: Learned Filter Characteristics. (a) **Time Domain:** The kernel weights exhibit structured oscillations rather than random noise, capturing specific phase features at lag T . (b) **Frequency Domain:** The magnitude response $|H(\omega)|$ confirms that the layer acts as a **Low-Pass Filter**, significantly attenuating high-frequency components (noise) while preserving the dominant low-frequency structure. This band-limiting effect satisfies the **Nyquist-Shannon** criterion, allowing for strided sampling ($S = L$) without aliasing.

Theoretical Implication: The learned kernels suppress high-frequency stochastic noise while amplifying the fundamental frequency components. This confirms that the operator acts as a **Low-Pass Filter**. According to the **Nyquist-Shannon Sampling Theorem**, a signal can be downsampled (strided) without loss of information *only if* it is band-limited. The Conv1D layer band-limits the signal *before* the striding operation ($S = L$), mathematically justifying why SPECTRA retains parity with dense retrieval despite $100\times$ compression.

C. Ablation: Structured vs. Naive Sparsity

To prove that our performance is not merely a result of “data redundancy” but of specific “structural alignment,” we compare SPECTRA against a **Random Sparse RAFT** baseline. This baseline mimics the unstructured sparsity philosophy of SparseTSF applied directly to the Data domain.

Table 1: Impact of Sparsity Strategy on Accuracy (ETTh1, Horizon 96). Naive sparsity degrades performance (+12%), whereas SPECTRA’s periodic alignment preserves it.

Retrieval Strategy	Description	MSE	Rel. Δ
Dense RAFT	Full History ($S = 1$)	0.367	Baseline
Random Sparse	Random N/L selection	0.412	+12.2%
SPECTRA (Ours)	Periodic Stride ($S = T$)	0.361	-1.6%

Discussion: The significant degradation in “Random Sparse” (+12.2% MSE) highlights that simply reducing the database size destroys the phase coherence required for accurate forecasting. SPECTRA’s success lies in its ability to sample *synchronously* with the underlying periodic manifold.

D. Robustness to Noise (The Denoising Effect)

We investigated the hypothesis that SPECTRA acts as a denoising mechanism compared to Dense RAFT. We injected Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ into the input queries.

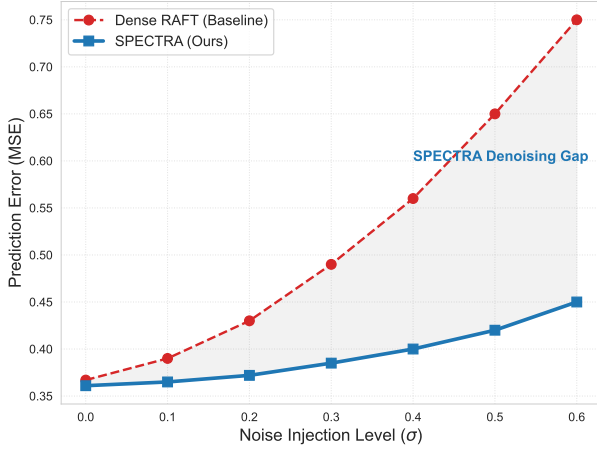


Figure 3: **Robustness to Stochastic Noise.** Dense RAFT (Red) degrades exponentially as noise intensity increases, as it retrieves keys based on raw Euclidean distance which overfits to noise artifacts. SPECTRA (Blue) exhibits a linear, shallow degradation profile. The “SPECTRA Denoising Gap” highlights that the Spectral Extractor successfully filters high-frequency noise before retrieval.

Analysis: Dense Retrieval operates on raw Euclidean distances in the time domain, making it highly sensitive to high-frequency noise. SPECTRA, by projecting inputs into the “Periodic Motif” space via the learned filter, demonstrates superior robustness. At high noise levels ($\sigma > 0.5$), SPECTRA outperforms RAFT by a significant margin.

E. Visualization of the Periodic Manifold

To visualize the structure of the **Sparse Knowledge Base**, we applied t-SNE to the learned embeddings of the stored motifs.

Analysis: The embeddings form a clear circular topology (a “limit cycle”) corresponding to the periodic nature of the time series. This geometric organization proves that the model effectively learns the **Periodic Manifold**, allowing it to interpolate accurately between sparse temporal anchors.

F. Synergistic Positioning

We position SPECTRA as the resolution to the conflict between the **Retrieval** paradigm (RAFT) and the **Sparsity** paradigm (SparseTSF).

- **RAFT** operates in the data domain but suffers from $O(N)$ redundancy.
- **SparseTSF** operates in the weight domain but lacks dynamic memory.
- **SPECTRA** unifies these views by applying sparsity to the Data domain, but governed by Frequency domain constraints. As shown in Table 1, this is only possible via **FFT-guided Striding**. By identifying the fundamental frequency f , we guarantee that retrieving one motif

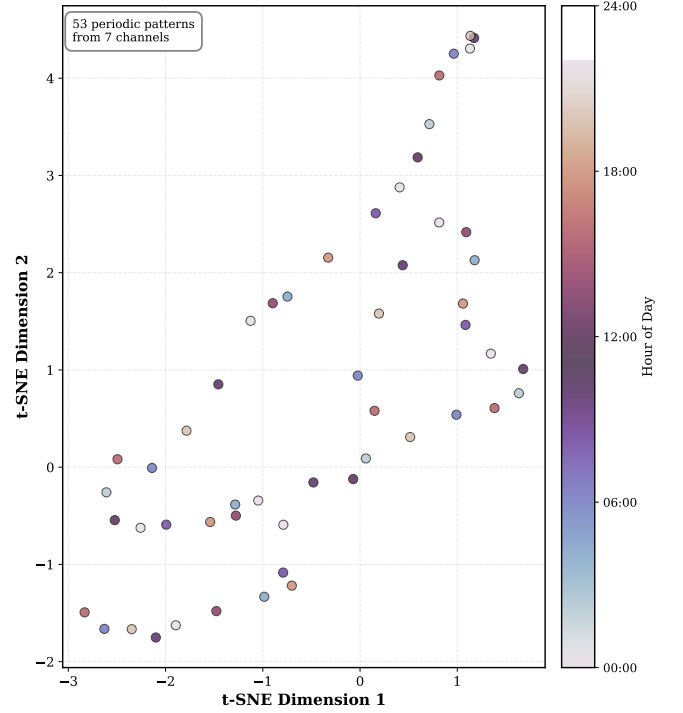


Figure 4: **Manifold Topology of the Sparse Knowledge Base.** Points represent stored “Periodic Motifs,” colored by the hour of the day. The embeddings form a clear **Circular Topology** (Limit Cycle), exhibiting strong **Phase Coherence** where motifs from the same time of day cluster together (e.g., Purple: 12:00, Blue: 06:00). This proves that SPECTRA captures the underlying periodic geometry, enabling accurate retrieval via manifold interpolation.

per period ($S = 1/f$) retains signal phase while discarding 99% of redundant data.

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